

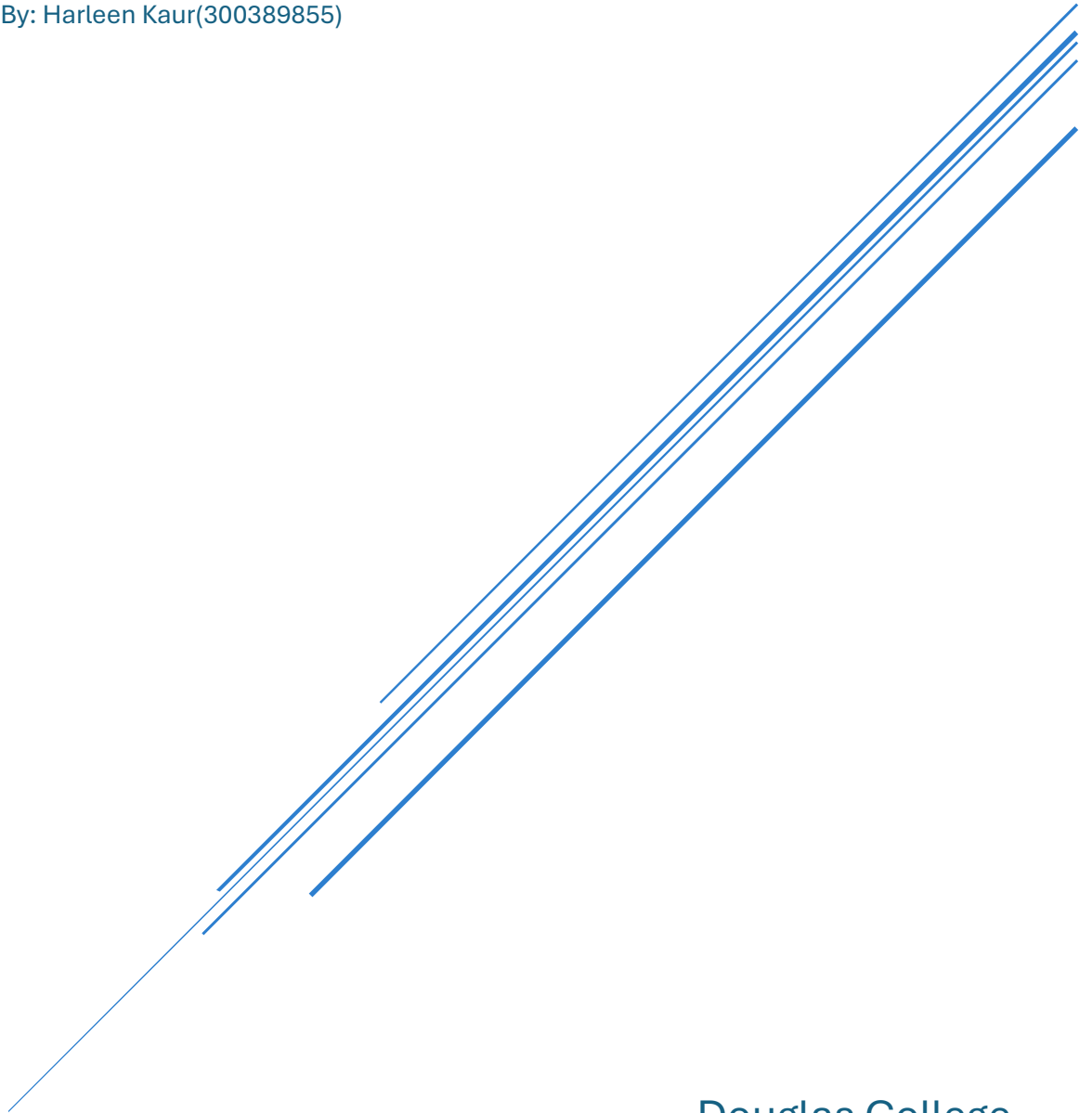
COMMUNITY EMERGENCY RESPONSE APP (CERA)

Applied Research Project

Report 2

Submitted to: Bambang Sarif

Submitted By: Harleen Kaur(300389855)



Douglas College
CSIS 4495 – Fall 2025

Date	Task Completed	Hours Spent
27 - Sept-2025	Fetches basic frontend and backend structure setup by Dilpreet, studied the codebase, and installed all necessary dependencies.	3 Hours
29-Sept-2025	Worked on MongoDB setup, resolved connection errors related to IP address and environment variables, and successfully tested database connection.	3 Hours
1-Oct-2025	Had meeting with Dilpreet, discussed the problems I faced. Developed initial frontend UI screens for login and registration including role selection and location fetching functionality during signup.	5 Hours
3-Oct-2025	Created backend models for users, defined schema fields for authentication and roles, and set up routes for registration and login.	3.5 Hours
5-Oct-2025	Implemented backend authentication including password hashing, JWT token generation, and role-based access control; integrated approval workflow for volunteer accounts.	3.5 Hours
07-Oct-2025	Handled edge cases for registration and login, fixed errors in route handling, and integrate frontend and backend together to make sure it works properly.	4 Hours
08-Oct-2025	Final testing of authentication refined logic, and update it on git hub.	3 Hours

DESCRIPTION OF WORK DONE:

Week 4 (27 Sept – 3 Oct 2025)

During this week, I started by fetching the basic frontend and backend structure set up by Dilpreet, studying the codebase, and installing all the necessary dependencies. I then worked on setting up MongoDB, resolved connection errors related to IP addresses and environment variables, and successfully tested the database connection. I had a meeting with Dilpreet to discuss the issues I faced and developed the initial frontend UI screens for login and registration. This included implementing role selection and location fetching functionality during signup. Finally, I created the backend models for users, defined schema fields for authentication and roles, and set up routes for registration and login.

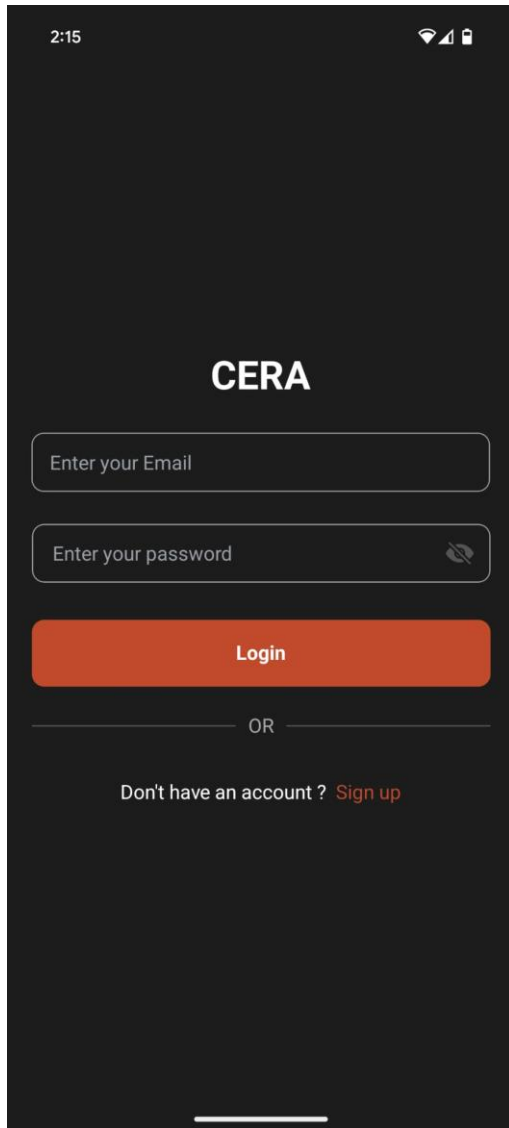
Week 5 (4 Oct – 8 Oct 2025)

In the second week, I implemented backend authentication, which included password hashing, JWT token generation, and role-based access control. I also integrated an approval workflow for volunteer accounts. I handled edge cases for registration and login, fixed errors in route handling, and integrated the frontend and backend to ensure proper functionality. Towards the end of the week, I performed final testing of authentication, refined the backend and frontend logic, and updated the project on GitHub.

Git Repo Check-in:

- **F2025_4495_050_HKa855/backend/models/User.js**
Mongoose schema for User with roles (resident, volunteer, coordinator)
- **F2025_4495_050_HKa855/backend/middleware/authMiddleware.js**
JWT verification and role-based authorization middleware (verifyToken, isCoordinator) to secure routes.
- **F2025_4495_050_HKa855/backend/routes/auth.js**
Authentication routes for registration and login.
- **F2025_4495_050_HKa855/backend/routes/users.js**
User routes
- **F2025_4495_050_HKa855/backend/server.js**
Express server setup, MongoDB connection, route integration, CORS, and JSON parsing
- **F2025_4495_050_HKa855/app/Mobile/cera-mobile/auth/signup.tsx**
Signup screen: username, email, phone, password, skills, role, and location inputs. Integrates with backend API for registration. Handles location permission
- **F2025_4495_050_HKa855/app/Mobile/cera-mobile/auth/login.tsx**
Login screen: email/password input, default coordinator login, volunteer approval check, error handling, navigation after login.
- **F2025_4495_050_HKa855/app/Mobile/cera-mobile/auth/_layout.tsx**
Stack navigator layout for authentication screens.

- **F2025_4495_050_HKa855/app/Mobile/cera-mobile/constants/config.ts**
App configuration constants, including API_BASE_URL for backend integration.



2:15

CERA

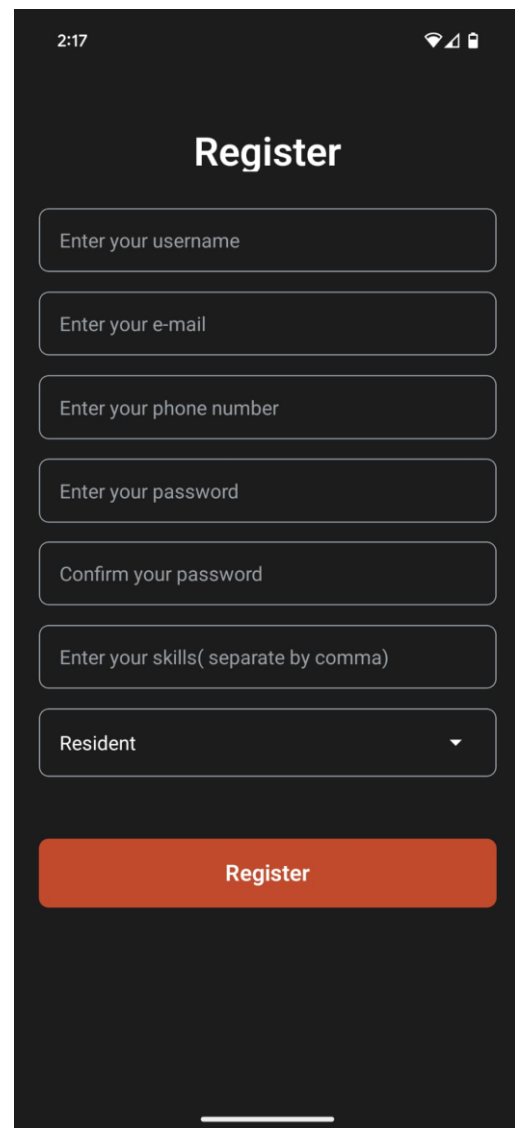
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Enter your password

Login

OR

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2:17

Register

Enter your username

Enter your e-mail

Enter your phone number

Enter your password

Confirm your password

Enter your skills(separate by comma)

Resident

Register

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\User\OneDrive - Douglas College\Desktop\Major Project\F2025_4495_050_HKa855\backend> node server.js
[dotenv@17.2.3] injecting env (3) from .env -- tip: ⚙️ load multiple .env files with { path: ['.env.local', '.env'] }
Connected to MongoDB successfully
Server running on http://localhost:5000
█
```

